Constrained Optimization Algorithms

Cong Ma

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Setup: stationarity and projection

Constrained Optimization Problems

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \in \mathcal{C} \end{aligned}$$

- $f(\cdot)$: convex function
- $\mathcal{C} \subseteq \mathbb{R}^n$: closed convex set

Stationary Points for Constrained Problems

Let $\mathcal{C} \subseteq \mathbb{R}^d$ be a closed, convex set and let $f : \mathcal{C} \to \mathbb{R}$ be continuously differentiable.

We consider the constrained problem

$$\text{(P)} \qquad \min_{\boldsymbol{x} \in \mathcal{C}} \; f(\boldsymbol{x}).$$

Definition 1 (Stationary point)

A point $\boldsymbol{x}^\star \in \mathcal{C}$ is called a stationary point of (P) if

$$\nabla f(\boldsymbol{x}^\star)^\top (\boldsymbol{x} - \boldsymbol{x}^\star) \;\geq\; 0, \qquad \forall \boldsymbol{x} \in \mathcal{C}.$$

- Interpretation: no feasible descent direction.

Stationarity Is Necessary for Local Optimality

Let $\mathcal{C} \subseteq \mathbb{R}^d$ be closed and convex, and let $f : \mathcal{C} \to \mathbb{R}$ be continuously differentiable. Consider

$$\text{(P)} \qquad \min_{\boldsymbol{x} \in \mathcal{C}} f(\boldsymbol{x}).$$

Theorem 2 (Stationarity as a necessary condition)

If $\boldsymbol{x}^\star$ is a local minimum of (P), then $\boldsymbol{x}^\star$ is a stationary point of (P), i.e.

$$\nabla f(\boldsymbol{x}^\star)^\top (\boldsymbol{x} - \boldsymbol{x}^\star) \;\geq\; 0, \qquad \forall \boldsymbol{x} \in \mathcal{C}.$$

For Convex Problems, Stationary $\Leftrightarrow$ Optimal

Let $\mathcal{C} \subseteq \mathbb{R}^d$ be closed and convex, and let $f : \mathcal{C} \to \mathbb{R}$ be continuously differentiable and convex. Consider

$$\text{(P)} \qquad \min_{\boldsymbol{x} \in \mathcal{C}} f(\boldsymbol{x}).$$

Theorem 3 (Stationarity is necessary and sufficient)

A point $\boldsymbol{x}^\star \in \mathcal{C}$ is stationary for (P), i.e.

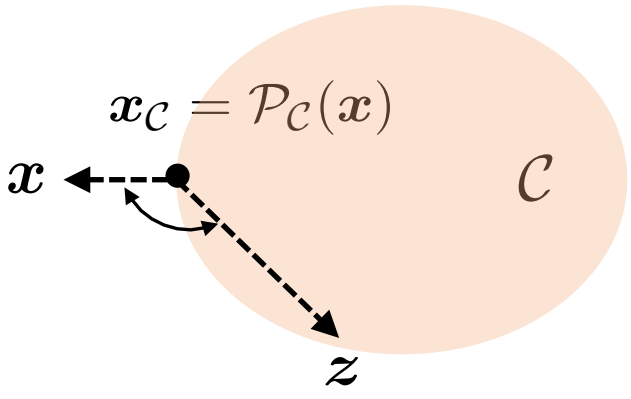
$$\nabla f(\boldsymbol{x}^\star)^\top (\boldsymbol{x} - \boldsymbol{x}^\star) \geq 0 \quad \forall \boldsymbol{x} \in \mathcal{C},$$

if and only if $\boldsymbol{x}^\star$ is a global optimal solution of (P).

Projection onto Closed and Convex Set

Let $\mathcal{C} \subseteq \mathbb{R}^d$ be nonempty, closed and convex. Given $\boldsymbol{x} \in \mathbb{R}^d$, define the projection

$$\boldsymbol{x}_{\mathcal{C}} := P_{\mathcal{C}}(\boldsymbol{x}) \in \arg\min_{\boldsymbol{z} \in \mathcal{C}} \ \|\boldsymbol{z} - \boldsymbol{x}\|_2^2.$$

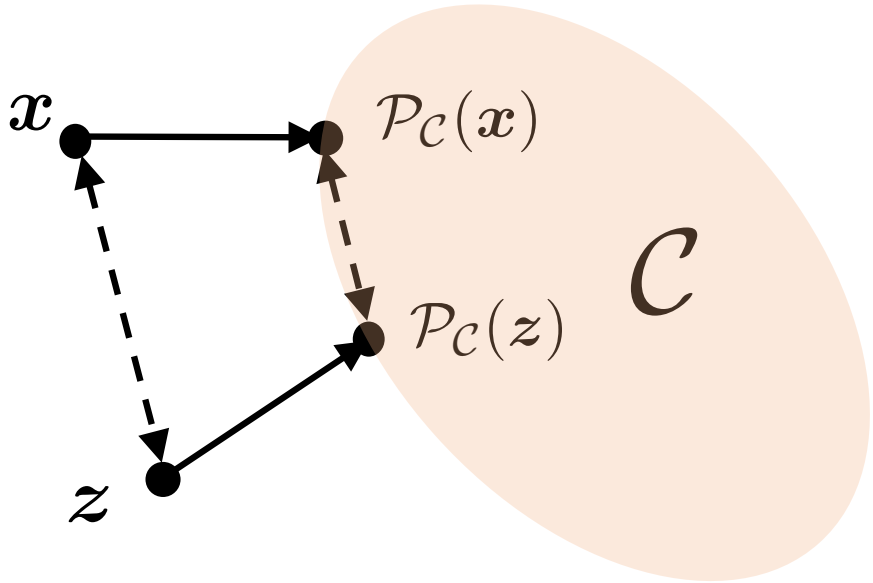


Fact 4 (Projection theorem)

Let $\mathcal{C}$ be closed & convex. Then $\boldsymbol{x}_{\mathcal{C}}$ is the projection of $\boldsymbol{x}$ onto $\mathcal{C}$ iff

$$(\boldsymbol{x} - \boldsymbol{x}_{\mathcal{C}})^\top (\boldsymbol{z} - \boldsymbol{x}_{\mathcal{C}}) \leq 0, \qquad \forall \boldsymbol{z} \in \mathcal{C}$$

Aside: Nonexpansiveness of Projection Operator



Fact 5 (Nonexpansiveness of projection)

For any $\boldsymbol{x}$ and $\boldsymbol{z}$, one has $\|\mathcal{P}_{\mathcal{C}}(\boldsymbol{x}) - \mathcal{P}_{\mathcal{C}}(\boldsymbol{z})\|_2 \leq \|\boldsymbol{x} - \boldsymbol{z}\|_2$

Projection Is Firmly Nonexpansive

Let $\mathcal{C} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex. Denote the projection by

$$P_{\mathcal{C}}(\boldsymbol{v}) := \arg\min_{\boldsymbol{x} \in \mathcal{C}} \|\boldsymbol{x} - \boldsymbol{v}\|_2.$$

Theorem 6

For any $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^d$:

1. ***(Firm nonexpansiveness)***

$$\big(P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\big)^\top (\boldsymbol{v} - \boldsymbol{w}) \;\geq\; \|P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\|_2^2. \tag{9.9}$$

2. ***(Nonexpansiveness)***

$$\|P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\|_2 \;\leq\; \|\boldsymbol{v} - \boldsymbol{w}\|_2. \tag{9.10}$$

Key Tool: Projection Theorem (Theorem 9.8)

Recall: $\boldsymbol{y} = P_{\mathcal{C}}(\boldsymbol{x})$ iff

$$(\boldsymbol{x} - \boldsymbol{y})^\top (\boldsymbol{z} - \boldsymbol{y}) \leq 0, \qquad \forall \boldsymbol{z} \in \mathcal{C}. \tag{9.11}$$

We will apply (9.11) twice, with the choices:

$$(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}) = (\boldsymbol{v},\, P_{\mathcal{C}}(\boldsymbol{v}),\, P_{\mathcal{C}}(\boldsymbol{w})) \quad \text{and} \quad (\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}) = (\boldsymbol{w},\, P_{\mathcal{C}}(\boldsymbol{w}),\, P_{\mathcal{C}}(\boldsymbol{v})).$$

Step 1: Two Inequalities from (9.11)

Substitute $\boldsymbol{x} = \boldsymbol{v}$, $\boldsymbol{y} = P_{\mathcal{C}}(\boldsymbol{v})$, $\boldsymbol{z} = P_{\mathcal{C}}(\boldsymbol{w})$ into (9.11):

$$\big(\boldsymbol{v} - P_{\mathcal{C}}(\boldsymbol{v})\big)^{\top}\big(P_{\mathcal{C}}(\boldsymbol{w}) - P_{\mathcal{C}}(\boldsymbol{v})\big) \leq 0. \tag{9.12}$$

Substitute $\boldsymbol{x} = \boldsymbol{w}$, $\boldsymbol{y} = P_{\mathcal{C}}(\boldsymbol{w})$, $\boldsymbol{z} = P_{\mathcal{C}}(\boldsymbol{v})$ into (9.11):

$$\big(\boldsymbol{w} - P_{\mathcal{C}}(\boldsymbol{w})\big)^{\top}\big(P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\big) \leq 0. \tag{9.13}$$

Step 2: Add Them and Rearrange to Get (9.9)

Add (9.12) and (9.13). Noting that

$$\big(P_{\mathcal{C}}(\boldsymbol{w}) - P_{\mathcal{C}}(\boldsymbol{v})\big) = -\big(P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\big),$$

we obtain

$$\big(P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\big)^{\top}(\boldsymbol{v} - \boldsymbol{w}) \;\geq\; \|P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\|_2^2,$$

which is exactly (9.9). $\square$

Step 3: Deduce Nonexpansiveness (9.10) from (9.9)

If $P_{\mathcal{C}}(\boldsymbol{v}) = P_{\mathcal{C}}(\boldsymbol{w})$, then (9.10) is trivial.

Otherwise, apply Cauchy–Schwarz:

$$\big(P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\big)^{\top}(\boldsymbol{v} - \boldsymbol{w}) \leq \|P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\|_2 \, \|\boldsymbol{v} - \boldsymbol{w}\|_2.$$

Combine with (9.9):

$$\|P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\|_2^2 \leq \|P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\|_2 \, \|\boldsymbol{v} - \boldsymbol{w}\|_2.$$

Divide by $\|P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\|_2 > 0$ to get

$$\|P_{\mathcal{C}}(\boldsymbol{v}) - P_{\mathcal{C}}(\boldsymbol{w})\|_2 \leq \|\boldsymbol{v} - \boldsymbol{w}\|_2,$$

which is (9.10). $\square$

Projected gradient method

Stationarity $\Leftrightarrow$ Projected-Gradient Fixed Point

Let $\mathcal{C} \subseteq \mathbb{R}^d$ be closed and convex, and let $f : \mathcal{C} \to \mathbb{R}$ be continuously differentiable. Fix any stepsize $s > 0$.

Consider the constrained problem

$$\text{(P)} \qquad \min_{\boldsymbol{x} \in \mathcal{C}} f(\boldsymbol{x}).$$

Theorem 7

A point $\boldsymbol{x}^\star \in \mathcal{C}$ is a stationary point of (P), i.e.

$$\nabla f(\boldsymbol{x}^\star)^\top (\boldsymbol{x} - \boldsymbol{x}^\star) \geq 0 \quad \forall \boldsymbol{x} \in \mathcal{C},$$

if and only if

$$\boldsymbol{x}^\star = P_{\mathcal{C}}\big(\boldsymbol{x}^\star - s\nabla f(\boldsymbol{x}^\star)\big). \tag{9.14}$$

Proof: Apply the Projection Theorem to $x^\star - s\nabla f(x^\star)$

Recall the projection theorem (Theorem 9.8):

$$\boldsymbol{y} = P_{\mathcal{C}}(\boldsymbol{x}) \quad \Longleftrightarrow \quad (\boldsymbol{x} - \boldsymbol{y})^\top (\boldsymbol{z} - \boldsymbol{y}) \leq 0, \quad \forall \boldsymbol{z} \in \mathcal{C}.$$

Apply it with

$$\boldsymbol{x} = \boldsymbol{x}^\star - s\nabla f(\boldsymbol{x}^\star), \qquad \boldsymbol{y} = \boldsymbol{x}^\star.$$

Then

$$\boldsymbol{x}^\star = P_{\mathcal{C}}\big(\boldsymbol{x}^\star - s\nabla f(\boldsymbol{x}^\star)\big)$$

holds iff for all $\boldsymbol{z} \in \mathcal{C}$,

$$\big((\boldsymbol{x}^\star - s\nabla f(\boldsymbol{x}^\star)) - \boldsymbol{x}^\star\big)^\top (\boldsymbol{z} - \boldsymbol{x}^\star) \leq 0.$$

Simplify: Fixed Point Condition $\Leftrightarrow$ Stationarity

The inequality becomes

$$(-s\nabla f(\boldsymbol{x}^\star))^\top(\boldsymbol{z}-\boldsymbol{x}^\star) \leq 0 \qquad \forall \boldsymbol{z} \in \mathcal{C}.$$

Since $s > 0$, divide by s and multiply by -1:

$$\nabla f(\boldsymbol{x}^\star)^\top(\boldsymbol{z}-\boldsymbol{x}^\star) \geq 0 \qquad \forall \boldsymbol{z} \in \mathcal{C}.$$

This is exactly the stationarity condition. $\square$

Remark: The Condition Does *not* Really Depend on s

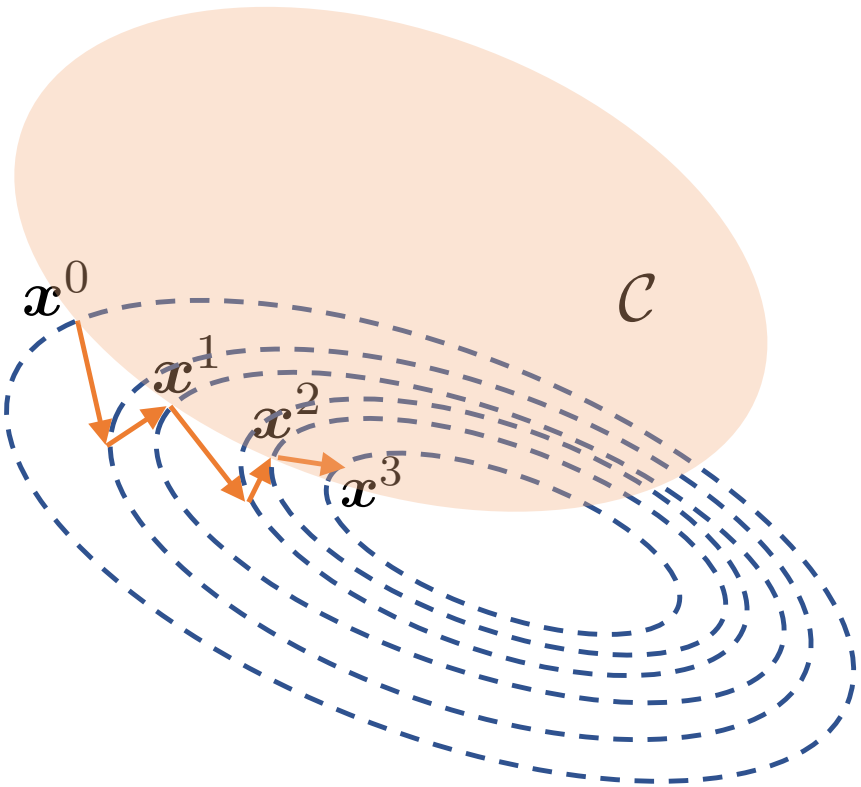
Although (9.14) is written with a stepsize $s > 0$, the equivalence shows:

$$\boldsymbol{x}^\star \text{ stationary} \quad \Longleftrightarrow \quad \boldsymbol{x}^\star = P_{\mathcal{C}}(\boldsymbol{x}^\star - s\nabla f(\boldsymbol{x}^\star)) \;\; \text{for } \textit{any } s > 0.$$

- This is the fixed-point form behind the projected gradient method:
$$\boldsymbol{x}^{t+1} = P_{\mathcal{C}}\big(\boldsymbol{x}^t - s\nabla f(\boldsymbol{x}^t)\big).$$
- Stationary points are exactly the fixed points of this iteration.

Convergence of projected gradient

Projected gradient descent



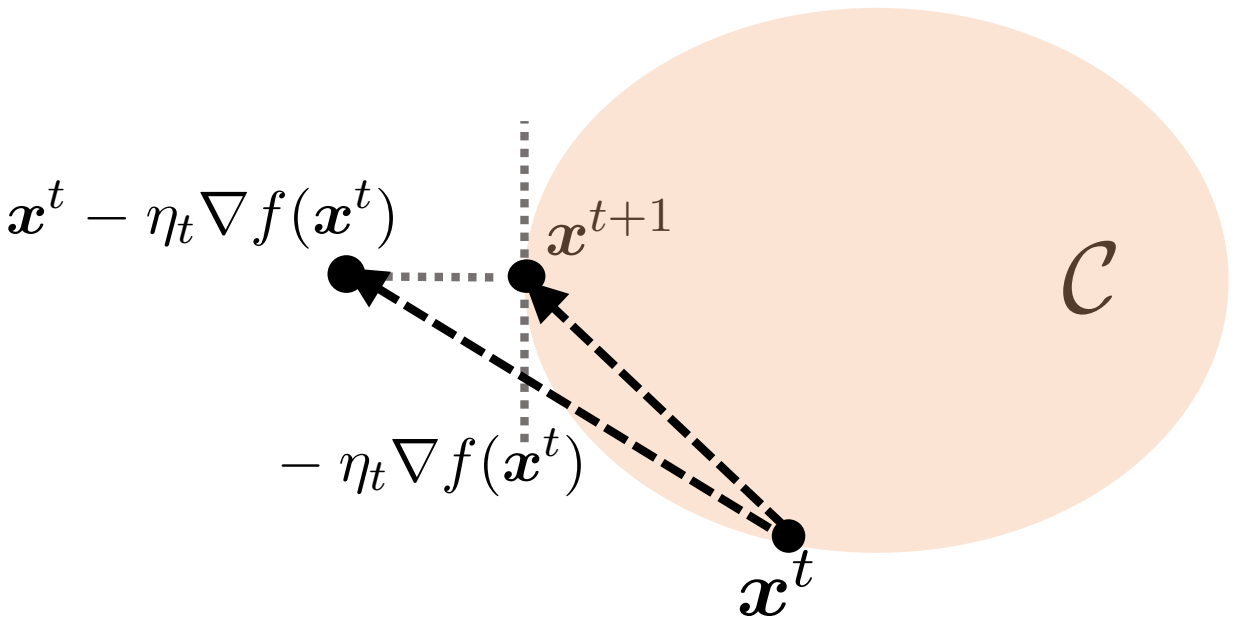
works well if projection onto $\mathcal{C}$ can be computed efficiently

for $t = 0, 1, \ldots$:

$$\boldsymbol{x}^{t+1} = \mathcal{P}_{\mathcal{C}}(\boldsymbol{x}^t - \eta_t \nabla f(\boldsymbol{x}^t))$$

where $\mathcal{P}_{\mathcal{C}}(\boldsymbol{x}) := \arg\min_{\boldsymbol{z} \in \mathcal{C}} \|\boldsymbol{x} - \boldsymbol{z}\|_2^2$ is $\underbrace{\text{Euclidean projection}}_{\text{quadratic minimization}}$ onto $\mathcal{C}$

Descent direction



From the above figure, we know

$$-\nabla f(\boldsymbol{x}^t)^\top (\boldsymbol{x}^{t+1} - \boldsymbol{x}^t) \geq 0$$

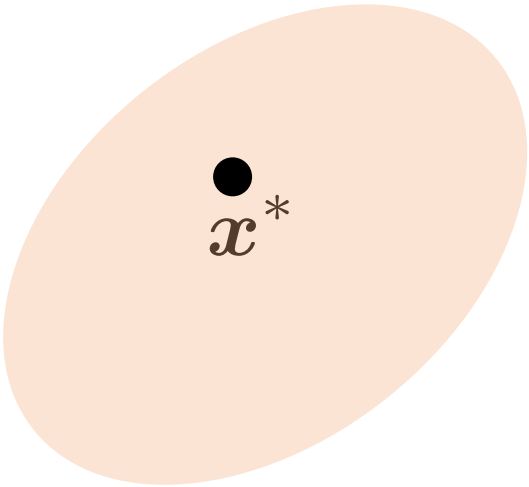
$\boldsymbol{x}^{t+1} - \boldsymbol{x}^t$ is positively correlated with the steepest descent direction

Strongly Convex and Smooth Problems

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \in \mathcal{C} \end{aligned}$$

- $f(\cdot)$: μ-strongly convex and L-smooth
- $\mathcal{C} \subseteq \mathbb{R}^n$: closed and convex

Convergence for Strongly Convex and Smooth Problems



Let's start with the simple case where $\boldsymbol{x}^*$ lies in the interior of $\mathcal{C}$ (so $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$).

Convergence for Strongly Convex and Smooth Problems

Theorem 8

Suppose $\boldsymbol{x}^* \in \mathrm{int}(\mathcal{C})$*, and let* f *be* μ*-strongly convex and* L*-smooth. If* $\eta_t = \frac{2}{\mu+L}$*, then*

$$\|\boldsymbol{x}^t - \boldsymbol{x}^*\|_2 \leq \left(\frac{\kappa-1}{\kappa+1}\right)^t \|\boldsymbol{x}^0 - \boldsymbol{x}^*\|_2$$

where $\kappa = L/\mu$ *is condition number*

- the same convergence rate as for the unconstrained case

Proof of Theorem 8

We have shown for the unconstrained case with $\nabla f(\boldsymbol{x}^*) = 0$ that

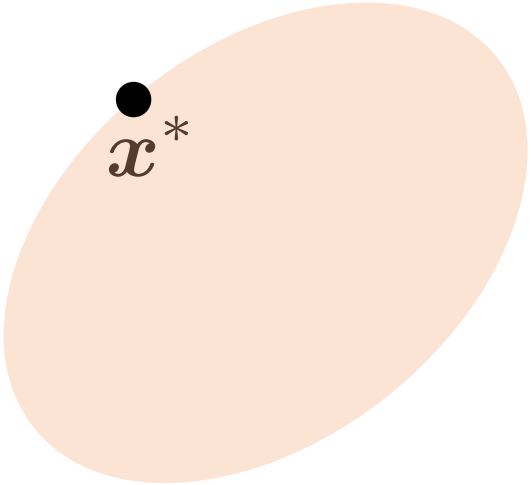
$$\|\boldsymbol{x}^t - \eta_t \nabla f(\boldsymbol{x}^t) - \boldsymbol{x}^*\|_2 \leq \frac{\kappa - 1}{\kappa + 1} \|\boldsymbol{x}^t - \boldsymbol{x}^*\|_2$$

From the nonexpansiveness of $\mathcal{P}_{\mathcal{C}}$, we know

$$\begin{aligned}
\|\boldsymbol{x}^{t+1} - \boldsymbol{x}^*\|_2 &= \|\mathcal{P}_{\mathcal{C}}(\boldsymbol{x}^t - \eta_t \nabla f(\boldsymbol{x}^t)) - \mathcal{P}_{\mathcal{C}}(\boldsymbol{x}^*)\|_2 \\
&\leq \|\boldsymbol{x}^t - \eta_t \nabla f(\boldsymbol{x}^t) - \boldsymbol{x}^*\|_2 \\
&\leq \frac{\kappa - 1}{\kappa + 1} \|\boldsymbol{x}^t - \boldsymbol{x}^*\|_2
\end{aligned}$$

Apply it recursively to conclude the proof

Convergence for Strongly Convex and Smooth Problems



What happens if we do not know whether $\boldsymbol{x}^* \in \text{int}(\mathcal{C})$?

- Main issue: $\nabla f(\boldsymbol{x}^*)$ may not be $\mathbf{0}$ (so the prior analysis may fail).

Convergence for Strongly Convex and Smooth Problems

Theorem 9 (projected GD for strongly convex and smooth problems)

Let f be μ-strongly convex and L-smooth. If $\eta_t \equiv \eta = \frac{1}{L}$, then

$$\|\boldsymbol{x}^t - \boldsymbol{x}^*\|_2^2 \leq \left(1 - \frac{\mu}{L}\right)^t \|\boldsymbol{x}^0 - \boldsymbol{x}^*\|_2^2$$

- Slightly weaker convergence guarantee than Theorem 8.

Proof of Theorem 9: One-Step Contraction

Set $\eta = \frac{1}{L}$ and

$$\boldsymbol{x}^+ := P_{\mathcal{C}}(\boldsymbol{x} - \eta \nabla f(\boldsymbol{x})), \qquad g_{\mathcal{C}}(\boldsymbol{x}) := \frac{1}{\eta}(\boldsymbol{x} - \boldsymbol{x}^+) = L(\boldsymbol{x} - \boldsymbol{x}^+).$$

Then the update is $\boldsymbol{x}^{t+1} = \boldsymbol{x}^+ = (\boldsymbol{x}^t)^+$.

Goal: show the one-step contraction

$$\|\boldsymbol{x}^+ - \boldsymbol{x}^\star\|_2^2 \;\leq\; \left(1 - \frac{\mu}{L}\right) \|\boldsymbol{x} - \boldsymbol{x}^\star\|_2^2,$$

then apply it recursively with $\boldsymbol{x} = \boldsymbol{x}^t$.

Key inequality (from projection):

$$\langle \nabla f(\boldsymbol{x}) - g_{\mathcal{C}}(\boldsymbol{x}),\, \boldsymbol{x}^\star - \boldsymbol{x}^+ \rangle \;\leq\; 0. \qquad (\clubsuit)$$

(This is Fact 4 applied to $\boldsymbol{x} - \eta \nabla f(\boldsymbol{x})$ and $\boldsymbol{x}^+ = P_{\mathcal{C}}(\cdot)$.)

Two Ingredients + One Line of Algebra

(1) Smoothness (descent lemma) with $\eta = \frac{1}{L}$:

$$f(\boldsymbol{x}^+) \leq f(\boldsymbol{x}) + \langle \nabla f(\boldsymbol{x}), \boldsymbol{x}^+ - \boldsymbol{x} \rangle + \frac{L}{2}\|\boldsymbol{x}^+ - \boldsymbol{x}\|_2^2 = f(\boldsymbol{x}) - \frac{1}{2L}\|g_{\mathcal{C}}(\boldsymbol{x})\|_2^2. \tag{A}$$

(2) Strong convexity:

$$f(\boldsymbol{x}) - f(\boldsymbol{x}^\star) \leq \langle \nabla f(\boldsymbol{x}), \boldsymbol{x} - \boldsymbol{x}^\star \rangle - \frac{\mu}{2}\|\boldsymbol{x} - \boldsymbol{x}^\star\|_2^2. \tag{B}$$

Combine (A)+(B) and use $f(\boldsymbol{x}^+) \geq f(\boldsymbol{x}^\star)$:

$$0 \leq f(\boldsymbol{x}^+) - f(\boldsymbol{x}^\star) \leq \langle \nabla f(\boldsymbol{x}), \boldsymbol{x}^+ - \boldsymbol{x}^\star \rangle + \frac{1}{2L}\|g_{\mathcal{C}}(\boldsymbol{x})\|_2^2 - \frac{\mu}{2}\|\boldsymbol{x} - \boldsymbol{x}^\star\|_2^2. \tag{C}$$

Now apply (♣) to replace $\langle \nabla f(\boldsymbol{x}), \boldsymbol{x}^+ - \boldsymbol{x}^\star \rangle$ by $\langle g_{\mathcal{C}}(\boldsymbol{x}), \boldsymbol{x}^+ - \boldsymbol{x}^\star \rangle$ (projection only helps).

Finish: Regularity $\Rightarrow$ Contraction

From (C) and (♣):

$$\langle g_{\mathcal{C}}(\boldsymbol{x}), \boldsymbol{x}^+ - \boldsymbol{x}^\star \rangle \geq \frac{\mu}{2}\|\boldsymbol{x} - \boldsymbol{x}^\star\|_2^2 - \frac{1}{2L}\|g_{\mathcal{C}}(\boldsymbol{x})\|_2^2.$$

Since $\boldsymbol{x}^+ = \boldsymbol{x} - \frac{1}{L} g_{\mathcal{C}}(\boldsymbol{x})$, we have

$$\langle g_{\mathcal{C}}(\boldsymbol{x}), \boldsymbol{x}^+ - \boldsymbol{x}^\star \rangle = \langle g_{\mathcal{C}}(\boldsymbol{x}), \boldsymbol{x} - \boldsymbol{x}^\star \rangle - \frac{1}{L}\|g_{\mathcal{C}}(\boldsymbol{x})\|_2^2.$$

Therefore the regularity inequality holds:

$$\boxed{\langle g_{\mathcal{C}}(\boldsymbol{x}), \boldsymbol{x} - \boldsymbol{x}^\star \rangle \geq \frac{\mu}{2}\|\boldsymbol{x} - \boldsymbol{x}^\star\|_2^2 + \frac{1}{2L}\|g_{\mathcal{C}}(\boldsymbol{x})\|_2^2} \qquad \text{(R)}$$

Finish: Contraction Inequality

Finally,

$$\begin{aligned}
\|\boldsymbol{x}^+ - \boldsymbol{x}^\star\|_2^2 &= \left\|\boldsymbol{x} - \boldsymbol{x}^\star - \frac{1}{L} g_{\mathcal{C}}(\boldsymbol{x})\right\|_2^2 \\
&= \|\boldsymbol{x} - \boldsymbol{x}^\star\|_2^2 - \frac{2}{L}\langle g_{\mathcal{C}}(\boldsymbol{x}), \boldsymbol{x} - \boldsymbol{x}^\star\rangle + \frac{1}{L^2}\|g_{\mathcal{C}}(\boldsymbol{x})\|_2^2 \\
&\leq \|\boldsymbol{x} - \boldsymbol{x}^\star\|_2^2 - \frac{\mu}{L}\|\boldsymbol{x} - \boldsymbol{x}^\star\|_2^2 = \left(1 - \frac{\mu}{L}\right)\|\boldsymbol{x} - \boldsymbol{x}^\star\|_2^2,
\end{aligned}$$

where we used (R). Apply recursively to conclude. $\square$

Convex and Smooth Problems

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \in \mathcal{C} \end{aligned}$$

- $f(\cdot)$: convex and L-smooth
- $\mathcal{C} \subseteq \mathbb{R}^n$: closed and convex

Convergence for Convex and Smooth Problems

Problem

$$\min_{\boldsymbol{x} \in \mathcal{C}} f(\boldsymbol{x}),$$

where f is convex and L-smooth and $\mathcal{C}$ is closed convex.

Projected Gradient Descent

$$\boldsymbol{x}^{t+1} = P_{\mathcal{C}}(\boldsymbol{x}^t - \eta \nabla f(\boldsymbol{x}^t)).$$

Theorem 10

Let $\eta \leq 1/L$ and let $\boldsymbol{x}^\star \in \arg\min_{\boldsymbol{x} \in \mathcal{C}} f(\boldsymbol{x})$. Then

$$f(\boldsymbol{x}^T) - f(\boldsymbol{x}^\star) \leq \frac{\|\boldsymbol{x}^0 - \boldsymbol{x}^\star\|^2}{2\eta T}.$$

Proof Idea

Let $\boldsymbol{g}^t = \nabla f(\boldsymbol{x}^t)$ and define

$$\boldsymbol{y}^t = \boldsymbol{x}^t - \eta \boldsymbol{g}^t, \qquad \boldsymbol{x}^{t+1} = P_{\mathcal{C}}(\boldsymbol{y}^t).$$

1. Projection geometry

For any $\boldsymbol{u} \in \mathcal{C}$,

$$\langle \boldsymbol{y}^t - \boldsymbol{x}^{t+1}, \boldsymbol{u} - \boldsymbol{x}^{t+1} \rangle \leq 0,$$

which implies

$$\eta \langle \boldsymbol{g}^t, \boldsymbol{x}^{t+1} - \boldsymbol{u} \rangle \leq \frac{1}{2}\Big(\|\boldsymbol{x}^t - \boldsymbol{u}\|^2 - \|\boldsymbol{x}^{t+1} - \boldsymbol{u}\|^2 - \|\boldsymbol{x}^{t+1} - \boldsymbol{x}^t\|^2 \Big).$$

2. Convexity

$$f(\boldsymbol{x}^t) - f(\boldsymbol{u}) \leq \langle \boldsymbol{g}^t, \boldsymbol{x}^t - \boldsymbol{u} \rangle.$$

3. Smoothness ($\eta \leq 1/L$)

$$f(\boldsymbol{x}^{t+1}) \leq f(\boldsymbol{x}^t) + \langle \boldsymbol{g}^t, \boldsymbol{x}^{t+1} - \boldsymbol{x}^t \rangle + \frac{1}{2\eta} \|\boldsymbol{x}^{t+1} - \boldsymbol{x}^t\|^2.$$

Finish the Proof

Adding the inequalities gives

$$f(\boldsymbol{x}^{t+1}) - f(\boldsymbol{u}) \leq \frac{1}{2\eta}(\|\boldsymbol{x}^t - \boldsymbol{u}\|^2 - \|\boldsymbol{x}^{t+1} - \boldsymbol{u}\|^2).$$

Choose $\boldsymbol{u} = \boldsymbol{x}^\star$ and sum for $t = 0, \ldots, T-1$:

$$\sum_{t=0}^{T-1} \left(f(\boldsymbol{x}^{t+1}) - f(\boldsymbol{x}^\star)\right) \leq \frac{\|\boldsymbol{x}^0 - \boldsymbol{x}^\star\|^2}{2\eta}.$$

Since $f(\boldsymbol{x}^t)$ decreases,

$$f(\boldsymbol{x}^T) - f(\boldsymbol{x}^\star) \leq \frac{1}{T}\sum_{t=0}^{T-1} \left(f(\boldsymbol{x}^{t+1}) - f(\boldsymbol{x}^\star)\right).$$

$$\boxed{f(\boldsymbol{x}^T) - f(\boldsymbol{x}^\star) \leq \frac{\|\boldsymbol{x}^0 - \boldsymbol{x}^\star\|^2}{2\eta T}}$$

Projection + Convexity + Smoothness $\Rightarrow$ Telescoping.

Frank-Wolfe algorithm

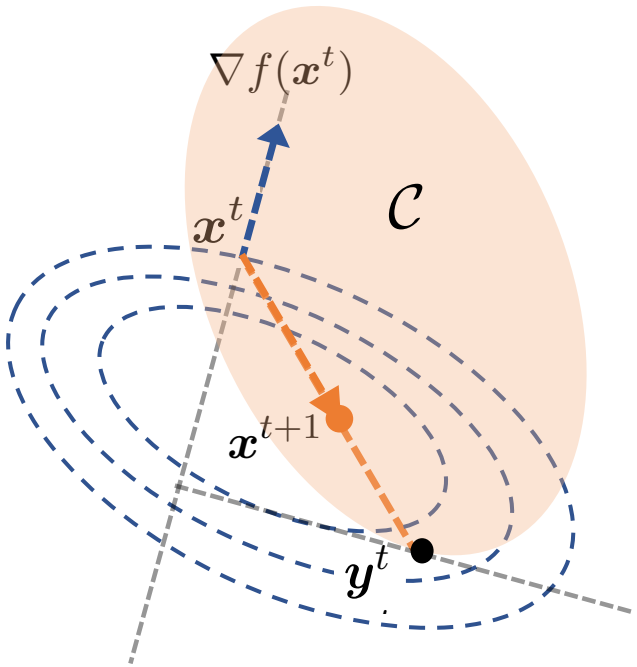
developed by Philip Wolfe and Marguerite Frank

Frank-Wolfe / Conditional Gradient Algorithm

Algorithm 1 Frank-wolfe (a.k.a. conditional gradient) algorithm

1: **for** $t = 0, 1, \ldots$ **do**
2: $\quad \boldsymbol{y}^t := \arg\min_{\boldsymbol{x} \in \mathcal{C}} \langle \nabla f(\boldsymbol{x}^t), \boldsymbol{x} \rangle$ (direction finding)
3: $\quad \boldsymbol{x}^{t+1} = (1 - \eta_t)\boldsymbol{x}^t + \eta_t \boldsymbol{y}^t$ (line search and update)

$$\boldsymbol{y}^t = \arg\min_{\boldsymbol{x} \in \mathcal{C}} \langle \nabla f(\boldsymbol{x}^t), \boldsymbol{x} - \boldsymbol{x}^t \rangle$$



Frank-Wolfe / Conditional Gradient Algorithm

Algorithm 2 Frank-wolfe (a.k.a. conditional gradient) algorithm

1: **for** $t = 0, 1, \ldots$ **do**
2: $\quad \boldsymbol{y}^t := \arg\min_{\boldsymbol{x} \in \mathcal{C}} \langle \nabla f(\boldsymbol{x}^t), \boldsymbol{x} \rangle$ (direction finding)
3: $\quad \boldsymbol{x}^{t+1} = (1 - \eta_t)\boldsymbol{x}^t + \eta_t \boldsymbol{y}^t$ (line search and update)

- main step: linearization of the objective function (equivalent to $f(\boldsymbol{x}^t) + \langle \nabla f(\boldsymbol{x}^t), \boldsymbol{x} - \boldsymbol{x}^t \rangle$)

  $\Longrightarrow$ linear optimization over a convex set

- appealing when linear optimization is cheap

- stepsize η_t determined by line search, or $\underbrace{\eta_t = \frac{2}{t+2}}_{\text{bias towards } \boldsymbol{x}^t \text{ for large } t}$

Two Ways to Enforce Constraints

Consider solving

$$\min_{\boldsymbol{x} \in \mathcal{C}} f(\boldsymbol{x})$$

Projected Gradient Descent

$$\boldsymbol{x}^{t+1} = \Pi_{\mathcal{C}}\big(\boldsymbol{x}^t - \eta \nabla f(\boldsymbol{x}^t)\big)$$

Requires solving

$$\boxed{\min_{\boldsymbol{x} \in \mathcal{C}} \|\boldsymbol{x} - \boldsymbol{z}\|_2^2}$$

- Euclidean projection
- quadratic optimization
- may be expensive

Frank–Wolfe

$$\boldsymbol{y}^t = \arg\min_{\boldsymbol{x} \in \mathcal{C}} \langle \nabla f(\boldsymbol{x}^t), \boldsymbol{x} \rangle$$

$$\boldsymbol{x}^{t+1} = (1 - \eta_t)\boldsymbol{x}^t + \eta_t \boldsymbol{y}^t$$

Requires solving

$$\boxed{\min_{\boldsymbol{x} \in \mathcal{C}} \langle \boldsymbol{g}, \boldsymbol{x} \rangle}$$

- linear optimization
- often much cheaper

Example: ℓ_1 Ball (Sparse Optimization)

Consider

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) \quad \text{s.t.} \quad \|\boldsymbol{x}\|_1 \leq R.$$

Projected Gradient Descent
Projection:

$$\Pi_{\|\cdot\|_1 \leq R}(\boldsymbol{z}) = \arg \min_{\|\boldsymbol{x}\|_1 \leq R} \|\boldsymbol{x} - \boldsymbol{z}\|_2^2.$$

Requires:

- sorting coordinates
- soft-thresholding
- cost $O(d \log d)$

Produces:

- dense vectors

Frank–Wolfe
Linear optimization:

$$\min_{\|\boldsymbol{x}\|_1 \leq R} \langle \boldsymbol{g}, \boldsymbol{x} \rangle.$$

Solution:

$$\boldsymbol{x} = -R \operatorname{sign}(g_i) \boldsymbol{e}_i, \quad i \in \arg\max_j |g_j|.$$

- find largest gradient entry
- cost $O(d)$

Produces:

- 1-sparse update

Example: Nuclear Norm Ball

Consider

$$\min_{\boldsymbol{X}} f(\boldsymbol{X}) \quad \text{s.t.} \quad \|\boldsymbol{X}\|_* \leq R$$

where $\|\boldsymbol{X}\|_*$ is the nuclear norm.

Projected Gradient Descent

Projection requires

$$\Pi_{\|\cdot\|_* \leq R}(\boldsymbol{Z}) = \arg\min_{\|\boldsymbol{X}\|_* \leq R} \|\boldsymbol{X} - \boldsymbol{Z}\|_F^2.$$

Requires full SVD

$$\boldsymbol{Z} = U \Sigma V^\top.$$

- compute all singular values
- cost: $O(n^3)$

Frank–Wolfe

Linear optimization:

$$\min_{\|\boldsymbol{X}\|_* \leq R} \langle \boldsymbol{G}, \boldsymbol{X} \rangle.$$

Solution:

$$\boldsymbol{X} = -R\, \boldsymbol{u}_1 \boldsymbol{v}_1^\top$$

where $(\boldsymbol{u}_1, \boldsymbol{v}_1)$ are top singular vectors of $\boldsymbol{G}$.

- only top singular vector
- cost: $O(n^2)$

Convergence for Convex and Smooth Problems

Theorem 11 (Frank-Wolfe for convex and smooth problems, Jaggi 13)

Let f be convex and L-smooth, and let $\mathcal{C}$ be compact and convex. With $\eta_t = \frac{2}{t+2}$,

$$f(\boldsymbol{x}^t) - f(\boldsymbol{x}^\star) \leq \frac{C}{t+2},$$

where

$$C := \max\left\{2Ld_{\mathcal{C}}^2,\ 2\big(f(\boldsymbol{x}^0) - f(\boldsymbol{x}^\star)\big)\right\},$$

and $d_{\mathcal{C}} = \sup_{\boldsymbol{x},\boldsymbol{y}\in\mathcal{C}} \|\boldsymbol{x} - \boldsymbol{y}\|_2$.

- for compact constraint sets, Frank–Wolfe achieves $O(1/t)$ convergence
- equivalently: ε-accuracy in $O(1/\varepsilon)$ iterations

Proof of Theorem 11

By smoothness,

$$
\begin{aligned}
f(\boldsymbol{x}^{t+1}) - f(\boldsymbol{x}^t) &\leq \nabla f(\boldsymbol{x}^t)^\top \Big(\underbrace{\boldsymbol{x}^{t+1} - \boldsymbol{x}^t}_{=\eta_t(\boldsymbol{y}^t - \boldsymbol{x}^t)}\Big) \\
&\quad + \frac{L}{2} \underbrace{\|\boldsymbol{x}^{t+1} - \boldsymbol{x}^t\|_2^2}_{=\eta_t^2 \|\boldsymbol{y}^t - \boldsymbol{x}^t\|_2^2 \leq \eta_t^2 d_{\mathcal{C}}^2} \\
&\leq \eta_t \nabla f(\boldsymbol{x}^t)^\top (\boldsymbol{y}^t - \boldsymbol{x}^t) + \frac{L}{2}\eta_t^2 d_{\mathcal{C}}^2 \\
&\leq \eta_t \nabla f(\boldsymbol{x}^t)^\top (\boldsymbol{x}^* - \boldsymbol{x}^t) + \frac{L}{2}\eta_t^2 d_{\mathcal{C}}^2 \quad \text{(since } \boldsymbol{y}^t \text{ is minimizer)} \\
&\leq \eta_t \big(f(\boldsymbol{x}^*) - f(\boldsymbol{x}^t)\big) + \frac{L}{2}\eta_t^2 d_{\mathcal{C}}^2 \qquad \text{(by convexity)}
\end{aligned}
$$

Letting $\Delta_t := f(\boldsymbol{x}^t) - f(\boldsymbol{x}^*)$ we get

$$
\Delta_{t+1} \leq (1 - \eta_t)\Delta_t + \frac{L d_{\mathcal{C}}^2}{2}\eta_t^2
$$

We then complete the proof by induction (which we omit here)

Finish the Proof

We obtained the recursion

$$\Delta_{t+1} \le (1-\eta_t)\Delta_t + \frac{Ld_{\mathcal{C}}^2}{2}\eta_t^2, \qquad \eta_t = \frac{2}{t+2}.$$

Lemma 12

$$\Delta_t \le \frac{C}{t+2}, \qquad C = \max\{2Ld_{\mathcal{C}}^2,\, 2\Delta_0\}.$$

Base case: By definition of C, $\Delta_0 \le C/2$.

Induction step: Assume $\Delta_t \le \frac{C}{t+2}$. Then

$$\begin{aligned}\Delta_{t+1} &\le \left(1 - \frac{2}{t+2}\right)\frac{C}{t+2} + \frac{Ld_{\mathcal{C}}^2}{2}\left(\frac{2}{t+2}\right)^2 \\ &\le \frac{C}{t+3} \qquad (\text{since } C \ge 2Ld_{\mathcal{C}}^2).\end{aligned}$$

Hence

$$f(\boldsymbol{x}^t) - f(\boldsymbol{x}^\star) \le \frac{C}{t+2}. \qquad \square$$

Duality Gap

Frank–Wolfe iterations admit a natural duality gap:

$$g_t := \nabla f(\boldsymbol{x}^t)^\top (\boldsymbol{x}^t - \boldsymbol{y}^t), \qquad \boldsymbol{y}^t = \arg\min_{\boldsymbol{s} \in \mathcal{C}} \langle \nabla f(\boldsymbol{x}^t), \boldsymbol{s} \rangle.$$

Lemma 13

$$g_t \;\geq\; f(\boldsymbol{x}^t) - f(\boldsymbol{x}^\star).$$

Proof.

By convexity,

$$f(\boldsymbol{s}) \geq f(\boldsymbol{x}^t) + \nabla f(\boldsymbol{x}^t)^\top (\boldsymbol{s} - \boldsymbol{x}^t) \qquad \forall \boldsymbol{s} \in \mathcal{C}.$$

Minimizing over $\boldsymbol{s} \in \mathcal{C}$ gives

$$f^\star \geq f(\boldsymbol{x}^t) + \min_{\boldsymbol{s} \in \mathcal{C}} \nabla f(\boldsymbol{x}^t)^\top (\boldsymbol{s} - \boldsymbol{x}^t) = f(\boldsymbol{x}^t) + \nabla f(\boldsymbol{x}^t)^\top (\boldsymbol{y}^t - \boldsymbol{x}^t).$$

Rearranging,

$$f(\boldsymbol{x}^t) - f(\boldsymbol{x}^\star) \leq \nabla f(\boldsymbol{x}^t)^\top (\boldsymbol{x}^t - \boldsymbol{y}^t) = g_t.$$

Strongly Convex Problems?

Can we hope to improve convergence guarantees of Frank-Wolfe in the presence of strong convexity?

- in general, NO
- maybe improvable under additional conditions

A Negative Result

Example:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x} + \boldsymbol{b}^\top \boldsymbol{x} \qquad (\boldsymbol{Q} \succ \boldsymbol{0}) \\ \text{subject to} \quad & \underbrace{\boldsymbol{x} = [\boldsymbol{a}_1, \cdots, \boldsymbol{a}_k]\boldsymbol{v},\ \boldsymbol{v} \geq \boldsymbol{0},\ \boldsymbol{1}^\top \boldsymbol{v} = 1}_{\boldsymbol{x} \in \text{convex-hull}\{\boldsymbol{a}_1, \cdots, \boldsymbol{a}_k\}} \qquad (=: \Omega) \end{aligned} \tag{1}$$

- suppose interior$(\Omega) \neq \emptyset$
- suppose the optimal point $\boldsymbol{x}^*$ lies on the boundary of Ω and is not an extreme point

A Negative Result

Theorem 14 (Canon & Cullum, '68)

Let $\{\boldsymbol{x}^t\}$ *be Frank-Wolfe iterates with exact line search for solving* (1)*. Then* $\exists$ *an initial point* $\boldsymbol{x}^0$ *s.t. for every* $\varepsilon > 0$*,*

$$f(\boldsymbol{x}^t) - f(\boldsymbol{x}^*) \geq \frac{1}{t^{1+\varepsilon}} \qquad \textit{for infinitely many } t$$

- example: choose $\boldsymbol{x}^0 \in \text{interior}(\Omega)$ obeying $f(\boldsymbol{x}^0) < \min_i f(\boldsymbol{a}_i)$
- in general, cannot improve $O(1/t)$ convergence guarantees

Positive Results?

To achieve faster convergence, one needs additional assumptions

- example: strongly convex feasible set $\mathcal{C}$
- active research topics

An Example of Positive Results

A set $\mathcal{C}$ is said to be μ-strongly convex if $\forall \lambda \in [0,1]$ and $\forall \boldsymbol{x}, \boldsymbol{z} \in \mathcal{C}$:

$$\mathcal{B}\Big(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{z},\ \frac{\mu}{2}\lambda(1-\lambda)\|\boldsymbol{x}-\boldsymbol{z}\|_2^2\Big) \subseteq \mathcal{C},$$

where $\mathcal{B}(\boldsymbol{a}, r) := \{\boldsymbol{y} \mid \|\boldsymbol{y}-\boldsymbol{a}\|_2 \le r\}$

- example: ℓ_2 ball

Theorem 15 (Levitin & Polyak '66)

Suppose f is convex and L-smooth, and $\mathcal{C}$ is μ-strongly convex. Suppose $\|\nabla f(\boldsymbol{x})\|_2 \ge c > 0$ for all $\boldsymbol{x} \in \mathcal{C}$. Then under mild conditions, Frank-Wolfe with exact line search converges linearly